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Randomness and statistics

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Spock : Random chance seems to have operated in our favor.

Dr. McCoy : In plain non-Vulcan English, we've been lucky.

Spock : I believe I said that, Doctor.

Star Trek original series, The Doomsday Machine

Problem: Is randomness a statistical notion, or rather an algorithmic notion?

Question: Can we define randomness based on frequency measures?

Answer: No. Some non-random numbers pass frequency tests.

One of the first achievements of Kolmogorov complexity, that we owe to Andrei Kolmogorov himself, is a definition of randomness. Previous attempts to define randomness were based on ideas such as predictability and statistical symmetry. A binary sequence is predictable if you can win money by betting on the next bit, knowing the past of the sequence. Basic statistical symmetry requires that all thinkable patterns with n instantiated bits should occur with the same frequency 2^{-n} . There should be as many 0s as 1s in the sequence, with frequency 0.5. There should be as many 00 as 11, with frequency 0.25. And the pattern $0^{*}11^{*}1$ should occur with frequency $1/16$.

You may use the program `RandomByHand.py` to measure the frequency of patterns in the set of sequences.

The program reads the file `RandomByHand.csv` that is supposed to contain hand-generated "random" sequences. You can compare with pseudo-random sequences generated by the program. (you may run `RandomByHand.py` several times to vary results for rare patterns). If the sequences contained in `RandomByHand.csv` were really generated by people trying to make them look random, you should see a difference.

Ming Li and Paul Vitányi showed in 1994 that a random binary sequence of length n should *not* contain runs of zeros longer than about $\log_2(n) + \log(\log(n))$. However, and contrary to intuition, a random binary sequence is still expected to contain long runs of zeros, up to about $\log_2(n) - \log(\log(n))$ if n is large enough.

What is random

0/1 point (ungraded)

Select all of the statements that are true.

☒ A random binary sequence of length 1000 should contain the sequence 101010



☒ A 1Mb random binary sequence should contain the first 10 binary digits of π

digits of π

☒

A 1Mb random binary sequence should contain the first 13 binary digits of π

☒

A black-and-white random image with 4000x4000 pixels should contain vertical white segments of 32 pixels.

✖

A 1000-bit random sequence will indeed contain 101010

A 1Mb random binary sequence should contain any predefined sequences up to 15 bits

A 16Mb random binary sequence is not expected to contain a predefined sequence larger than 29 bits

Submit

You have used 1 of 1 attempt

 Answers are displayed within the problem

In 1919, Richard von Mises proposed that a binary sequence should be regarded as random iff any sub-sequence extracted independently from the content has bit frequency $\frac{1}{2}$ at the limit. This idea seems to make sense.

Unfortunately, these frequency tests are obviously inadequate to capture the idea of randomness. The following number, known as the Champernowne constant, is "normal", which means that it passes all frequency tests.

$c_{10} =$
0.12345678910111213141516171819202122232425262728293031323334353
6...

The Champernowne constant is obtained by concatenating successive integers (you may generate c_b

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